

# Answers to the quiz on Compilers

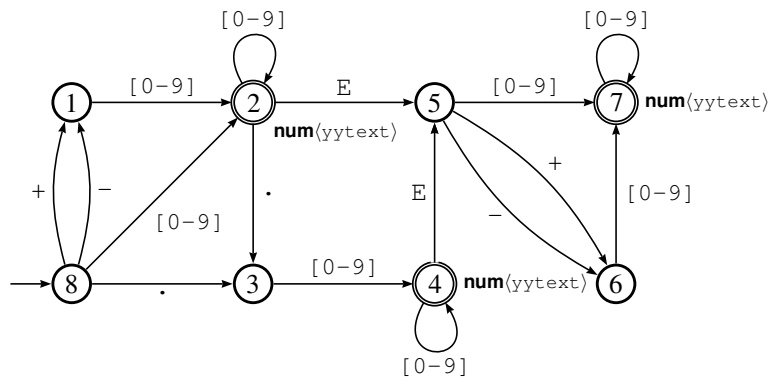
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**Question.** Consider the following Lex regular expression and propose a transition diagram which allows the recognition of the same lexemes as `num`.

`num (\.[0-9]+|[+\-]?[0-9]+(\.[0-9]+)?)(E[+\-]?[0-9]+)?`

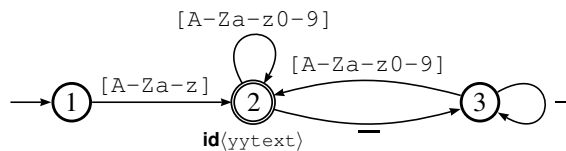
**Answer.**



**Question.** Consider the following Lex regular expression and propose a transition diagram which allows the recognition of the same lexemes as `id`.

`id [A-Za-z]([_]*[A-Za-z0-9])*`

**Answer.**



**Question.** Assume the following input buffer where `␣` represents a blank character.

`return␣␣x+␣.5E2+y_0`

Complete the following Lex skeleton

```
%{
#include<string.h>
char* keyword[] = {"else", "if", "return", "then"};
%}
num    (\.[0-9]+|[+\-]?[0-9]+(\.[0-9]+)?)(E[+\-]?[0-9]+)?
id     [A-Za-z]([_]*[A-Za-z0-9])*
```

```
%%
{num}  {
```

```
      }
{id}   {
```

```
      }
```

```
%%
```

such that the array `keyword` is used<sup>1</sup> and that the output is

```
kwd<return>
id<x>
plus<>
num<.5E2>
plus<>
id<y_0>
```

**Answer.**

```
%{
#include<string.h>
char* keyword[] = {"else", "if", "return", "then"};
%}
num    (\.[0-9]+|[+\-]?[0-9]+(\.[0-9]+)?)(E[+\-]?[0-9]+)?
id     [A-Za-z]([_]*[A-Za-z0-9])*
ws     [\n\t]+
%%
```

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<sup>1</sup>You may use the C function `int strcmp(const char *s1, const char *s2)` which returns 0 if the strings `s1` and `s2` are equal.

```

{num} { printf ("num<%s>\n",yytext); }
{id}  { int index = 0;
      while (index <= 3 && strcmp(keyword[index],yytext))
          index++;
      if (index == 4) printf("id<%s>\n",yytext);
      else printf("kwd<%s>\n",yytext);
      }
{ws}  {}
"+"   { printf ("plus<>\n"); }
%%

```

**Question.** Define the meaning of the pointers  $\downarrow$ ,  $\uparrow$  and  $\uparrow\uparrow$  presented in class and show how the input is analysed using the transition diagrams of the previous questions.

**Answer.** See handouts and previous answers.